Technical Report - Multi-Modal RAG QA System

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Submission: AI/ML Internship Assessment
Interface: Streamlit QA Application
Input Document: qatar_test_doc.pdf

Github link - https://github.com/skadesplaire01/Multi-Modal-RAG-QA-System.git

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1) Problem Statement

Modern LLM applications must answer questions from real-world documents, not only plain text. Complex reports (IMF Article IV type documents) contain text, tables, charts/figures, scanned images, and footnotes, where valuable insights may exist outside normal paragraphs.

This project implements a Multi-Modal Retrieval-Augmented Generation (RAG) pipeline that answers user queries using evidence retrieved from the document and provides page-level citations for trust and verification.

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2) System Architecture

Pipeline Overview : PDF → Ingestion → Chunking → Embeddings → Vector Index → Retrieval → (Rerank) → Answer + Citations

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3) Design Choices

- FAISS was selected for fast similarity search and scalable retrieval.
- Chunking with overlap (900/150) improves retrieval consistency for long paragraphs.
- CrossEncoder reranking (bonus) improves relevance when multiple chunks are similar.

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4) Core Components

- **Ingestion**: Extracts text, tables, and OCR content (Tesseract) from the PDF with page metadata.
- **Chunking:** Splits extracted content into retrieval-friendly chunks (chunk_size=900, overlap=150).
- **Embedding + Indexing:** Generates embeddings using SentenceTransformers and stores them in a FAISS vector index.
- **Retrieval + Reranking:** Uses Top-K retrieval and optional CrossEncoder reranking for better relevance.
- **QA Interface + Citations:** Streamlit UI provides answers with page-level citations and retrieved evidence display.

4) Tools & Tech Stack

Python ,Streamlit (UI), FAISS (vector search), SentenceTransformers (embeddings) , Tesseract OCR (image text extraction), PDF parsing utilities (text/table extraction)

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5) Demo Application (Streamlit)

The demo interface includes:

- Build/Rebuild index button
- Query input box
- Top-K retrieval slider
- Optional reranker toggle
- Answer, sources, and retrieved evidence preview

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6) Key Observations

- Table extraction improves accuracy for numeric queries like CPI inflation and projections.
- OCR increases coverage for figures/charts, but quality depends on image clarity.
- Reranking improves retrieval relevance when multiple chunks look similar, with a small latency increase.

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7) Output

For each user query, the system returns:

- **Final Answer:** Evidence-based response generated from retrieved document context
- **Source Attribution:** Page-level citations (page numbers) for traceability
- **Retrieved Evidence:** Top retrieved chunks displayed with metadata:
  - Page number
  - Chunk type: text / table / OCR
  - Similarity score / rerank score

This ensures the QA responses are transparent, verifiable, and grounded in the document**.**

Conclusion

This project demonstrates a complete multi-modal RAG pipeline capable of answering questions from a complex document using retrieved evidence and citations. The modular architecture enables future enhancements such as hybrid retrieval (RRF), richer evaluation dashboards, and improved summary generation.